

Worksheet 4

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1. Aim/Overview of the practical:

Interface an Arduino Uno, Led and Buzzer with RFID reader/writer MFRC522 and register the card's UID.

Write a program to check whether the card used is authorized or unauthorized.

- a) If the card is authorized, then Turn ON only "Yellow Led" and buzzer will Beep twice.
- b) If the card is unauthorized then Turn ON only "Blue Led" and buzzer will Beep three times.

2. Apparatus:

Hardware Requirements: Breadboard
MFRC 522 RFID Sensor
Buzzer
Jumper wires
LED (2 pieces)
Arduino Uno Board

Software requirements: Arduino IDE
Laptop

3. Circuit Diagram(TinkerCad):

Coding:

```
#include <SPI.h>
#include <MFRC522.h>
// RFID pins
#define SS_PIN 10
#define RST_PIN 9
// Output pins
#define YELLOW_LED 6
#define BLUE_LED 7
#define BUZZER 8
MFRC522 rfid(SS_PIN, RST_PIN);
byte authorizedUID[] = {0xA3, 0xB4, 0xC5, 0xD6};
```

```
void setup() {
  Serial.begin(9600);
  SPI.begin();
  rfid.PCD_Init();

  pinMode(YELLOW_LED, OUTPUT);
  pinMode(BLUE_LED, OUTPUT);
  pinMode(BUZZER, OUTPUT);

  digitalWrite(YELLOW_LED, LOW);
  digitalWrite(BLUE_LED, LOW);
  digitalWrite(BUZZER, LOW);

  Serial.println("RFID Reader Ready...");
  Serial.println("Place your RFID card near the reader.");
}

void loop() {
  // Check whether a new card is present
  if (!rfid.PICC_IsNewCardPresent()) {
    return;
  }
  // Read the card
  if (!rfid.PICC_ReadCardSerial()) {
    return;
  }
  // Print UID
  Serial.print("Card UID: ");

  for (byte i = 0; i < rfid.uid.size; i++) {
    if (rfid.uid.uidByte[i] < 0x10) {
      Serial.print("0");
    }
    Serial.print(rfid.uid.uidByte[i], HEX);
    Serial.print(" ");
  }
  Serial.println();
  // Check whether UID is authorized
  if (checkUID()) {
    Serial.println("AUTHORIZED CARD");
    // Yellow LED ON
    digitalWrite(YELLOW_LED, HIGH);
    digitalWrite(BLUE_LED, LOW);

    // Beep twice
    for (int i = 0; i < 2; i++) {
      digitalWrite(BUZZER, HIGH);
      delay(200);
    }
  }
}
```

```
digitalWrite(BUZZER, LOW);
delay(200);
}
delay(2000);
// Turn LED OFF
digitalWrite(YELLOW_LED, LOW);

} else {
Serial.println("UNAUTHORIZED CARD");
// Blue LED ON
digitalWrite(YELLOW_LED, LOW);
digitalWrite(BLUE_LED, HIGH);
// Beep three times
for (int i = 0; i < 3; i++) {
digitalWrite(BUZZER, HIGH);
delay(200);
digitalWrite(BUZZER, LOW);
delay(200);
}
delay(2000);
// Turn LED OFF
digitalWrite(BLUE_LED, LOW);
}
// Stop communication with current card
rfid.PICC_HaltA();
rfid.PCD_StopCrypto1();
delay(500);
}
bool checkUID() {

// First check UID length
if (rfid.uid.size != sizeof(authorizedUID)) {
return false;
}

// Compare every byte
for (byte i = 0; i < rfid.uid.size; i++) {

if (rfid.uid.uidByte[i] != authorizedUID[i]) {
return false;
}
}
return true;
}
```

5. Learning outcomes (What I have learnt):

- 1.RFID sensor and its components.
- 2.Interfacing of Arduino uno board with RFID sensor.
- 3.Differences between RFID card and reader, in the context of active and passive elements.